AUTHORIZATION			
	NAME AND TITLE	**SIGNATURE**	**DATE**
Issued by:	Yael Goldbrener, QA Manager	See M-Files	
Reviewed and Approved by:	Natan Blanks, SVP R&D	See M-Files	

REVISION HISTORY				
REV. #	**CO #**	**DESCRIPTION**	**PREPARED BY**	**DATE**
02.03	NA	Moved some detail from the Quality Manual to this SOP and from this SOP to the new work instructions relating to it.	Alexey Sorokin, Scrum Master	Jan 24th, 2011
03.00	011	QA-SOP-0002 Design and Development was changed and split to QMS-7.3.1, QMS-7.3.7, QMS-7.3.7 and QMS-7.3.8. Updates following periodic review and QMS transfer to the PeriGen Israel office have been included.	Tsvia Erlich, Regulatory Affairs Consultant	April 2nd, 2013
04.00	076	Updated to comply with ISO 13485:2016; added reporting of significant change considerations country-specific requirements	Sharon Resnick, QA Manager	August 20th, 2018
04.01	102	Added different sources for design changes, added reference to CAPA and Complaint procedure, replaced QA/RA Manager with Management Representative	Yael Goldbrener, QA Manager	Sep 18th, 2019
04.02	110	Added SVP Product Management to required approvers	Natan Blanks, GM Israel Operations	Feb 27, 2020
04.03	118	Minor corrections and clarifications, for harmonization with Cary	Natan Blanks, SVP R&D	April 27, 2020

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1. Purpose

The purpose of this procedure is to define the methods for initiating, reviewing, verifying and as appropriate validating, approving, and implementing changes in a design or product during:

1.1. The development process
1.2. After release to market

In order to ensure that such changes are implemented in a controlled and effective manner.

2. Scope

This procedure applies to:

2.1. All Design and development projects in, or for, PeriGen.
2.2. All changes made to products and production process, following the release of the product to market, in or for PeriGen.
2.3. This procedure **does not** apply to products that do not meet the definition of a medical device at the target market or product that defined as medical device but are exempted from quality system.

3. Applicable documents

Standards and Regulations:

ISO 13485:2016 – Medical Devices-Quality Management Systems-Requirements for regulatory purposes, section 7.3.9.
Health Canada Medical Devices Regulations, sections 1, 34, 43 (1) (b).
FDA Quality System Regulations – 21 CFR Part 820.30 (i).
FDA Deciding When to Submit a 510(k) for a Change to an Existing Device, October 25, 2017
FDA Deciding When to Submit a 510(k) for a Software Change to an Existing Device, October 25, 2017

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NBOG Guidance for manufacturers and Notified Bodies on reporting of Design Changes and Changes of the Quality System

Procedures and Forms:

FQMS-4.2.3-002- Document Change Order Form.
FQMS-7.3.7-001– ECO Control Log.
FQMS-7.3.7-002 – Design & Development Change Form.
FQMS-7.3.7-003 - Engineering Change Order Form (ECO).
FQMS-7.3.7-004 – Field Change Order (FCO) Form.
QMS-7.3.1 – Design and Development Planning.
QMS-7.3.2 – Design and Development Input (Requirements).
QMS-7.3.3 – Design Output, Design Verification and Design Validation.
QMS-7.3.4 – Design Review.
QMS-7.3.8 – Design Transfer and Product Release.
QMS-7.5.8– Unique Device Identification.
QMS-8.5.2– Corrective and Preventive Action (Improvement)
QMS-8.5.1 – Customer Support and Complaint Handling.

Work Instructions:

RD-WI-001 – Risk Management.
RD-WI-002 – Design History File.
SW-WI-001 – Software Development Life Cycle Procedure.
RA-WI-002– FDA Regulatory Requirements.
RA-WI-003– Canadian Regulatory Requirements.
RA-WI-005 – FDA Recall and Reporting Requirements.
RA-WI-006 – Canadian Recall and Reporting Requirements.

4. Definitions and Abbreviations

SW – Software.
DR – Design Review.
WI – Work Instruction.
ECR – SW Change Request.
ECO – SW change Order.
DHF – Design History File.
SPM – Software Project Manager.

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5. Procedure

5.1. Design change control applies to changes to inputs or outputs as a result of design verification or design validation, changes to labeling or packaging, changes to enhance a product's performance or correct a nonconformity, changes of production process(es), and changes that result from product complaints.

5.2. <u>The change process entails two stages:</u>

- Approval of a required product change.
- Updating of affected documents and change implementation.

5.3. <u>Changes to the design under design and development project</u>

5.3.1. Any proposed change in a design under development, following design freeze, but prior to market release, should be provided to the Development Project Manager using the **FQMS-7.3.7-002,** *Design & Development Change Form*. Software changes shall be handled according to **SW-WI-001,** *Software Development Life Cycle* procedure**.**

5.3.2. The initial review of the required change and its implication should be done by the Development Project Manager.

5.3.3. If the Development Project Manager approves the change then the request is handed to the SVP R&D , who is responsible to review and approve the proposed change, and to decide and document the extent of the design change review to be conducted (as described below).

5.3.4. In case of minor or limited change, the SVP R&D may decide to reduce the extent of the review, but under no circumstance, a design change will be implemented without a documented review.

5.3.5. The Review shall include the following considerations, as applicable according to the design phase:

- No adverse effect to safety, effectiveness or any design features.
- Impact of the change on the design input requirements, product definition, intended use or the regulatory status and requirements or additional activities, deliverables and other documents required.
- Software lifecycle activities and documents related to the change as defined in **SW-WI-001**, *Software Development Life Cycle* procedure.
- The need to update the risk analysis related to the impact of the proposed change.
- The need to conduct a DR to review the concept or the verification activities performed or to approve the change.

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- Other components or elements of the design, or associated with the design, affected by the change.
- Interfaces with external subcontractors and other groups in PeriGen (such as Clinical Team, Marketing, Customer Support, Service, Installation).
- The need to perform verification and validation activities (such as SW verification and validation, human factor testing, etc.) including the need to repeat approved verification and validation activities.
- Influence on the installation, use, service, user manual, user training and/ or labeling materials.
- The need to update the essential requirements as may be required by regulatory authorities.
- The need to report the change to regulatory authorities.
- Other, related to the specific change.

5.3.6. The Development Project Manager shall record all activities to be performed using **FQMS-7.3.7-002**, *Design & Development Change Form*. The following detailes are to be included: Activity, person responsible to execute the activity, documents and record to be updated (such as design specification, technical drawings, user manual, test protocols, etc.) and internal and external personnel to notify.

5.3.7. In case the changes involves change to the SW, The SPM (Software Project Manager) shall define and record the SW life cycle activites related to the change. The activites shall be recorded in the **FQMS-7.3.7-002**, *Design & Development Change Form*.

5.3.8. The Development Project Manager shall monitor the performance and the results of all required activities.

5.3.9. The activities performed and, where applicable, their results shall be approved by the SVP R&D, prior to the implementation of the change.

5.3.10. As applicable, design change Design Reviews will be conducted. Software design changes shall be conducted according to the requirements set in the **SW-WI-001,** *Software Development Life Cycle* procedure**.**

5.3.11. When a change is approved, the Development Project Manager shall ensure that all personnel involved in the development of the concerned design are properly updated with the nature and implication of the change including external subcontractors.

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5.4. Changes to product design following release to market

5.4.1. General

5.4.1.1. The process of implementing a change in product design or in its labeling, intended use or installation process is divided into the following sequential stages intended to:

- Determine the content and appoint of the reviewing team.
- Verify, and as appropriate, validate that the required effect of the change, on the product and on the user, is achieved and no unforeseen adverse effect may occur. This is done by review, evaluation analysis, calculations and actual testing.
- Approve, alter or reject the change, based on the above results.
- Determine the activities/ deliverables required for implementing the change.
- Verify completion of all activities/ deliverables.
- Implement change and release the prodct to market.

5.4.2. Change Proposal and Review

5.4.2.1. Any employee, subcontractor or third party is entitled to initiate a proposal for change in the product, its intended use, labeling or installation procedure. The proposal shall be documented by a company employee in the **FQMS-7.3.7-003,** *Engineering Change Order (ECO) Form* to include general details and change type. The form will be forwarded to the Product Owner for further handling.

5.4.2.2. The Product Owner shall provide a detailed description of Change and Objectives, as applicable.

5.4.2.3. The Management Representative or QA Manager shall review the change request to evaluate the required inputs to conduct the change by determining appropriate reviewers for the review of the requested change and for the determination of the required activities/ deliverables as part of the change implementation.

5.4.2.4. The reviewers shall include at least the SPM, Product Owner, SVP Product Management, and the Management Representative or QA Manager. Additional reviewers, as listed below in paragraph 5.4.2.5, are required according to the scope of the change and it's implication on the product performance, safety, interfaces with other devices, intended use, operation process and regulatory status.

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5.4.2.5. Additional reviewers, according to the nature of the change, may include: General Manager or CEO, Chief Product Officer, development team leader, V&V team leader, Marketing manager, Operations manager, Clinical manager, Support team manager, Regulatory consultants, Human Factors , CFO.

5.4.2.6. Whenever a change on a marketed device may raise safety or effectiveness issues, , or require new or revised regulatory submissions and/or notifications, a regulatory expert must be consulted.

5.4.2.7. The duties of the reviewers are to:

- Determine the activities for verifying, and as appropriate validating the impact of the change on the product and the user, each in his own field.
- Review the results of these activities.
- Determine the activities required for implementing the change.
- Provide recommendations related to the descision if to approve, alter or reject the change.

5.4.2.8. The change reviewers shall review all aspects of the change request, including the following, as applicable:

5.4.2.9. Activities related to the approval of the change (activities intended to verify that if the change will be implemented the safety and the effectiveness of the device will not be affected and no unforseen adverse effect will emerge):

- Conformance to product specification and customer requirements and product effectiveness.
- The need to perform/ update Risk Analysis.
- Software life cycle activities, including software Verification and Validation activities.
- Complience with all pertinent regulatory and standard requirements.
- Regulatory aspects of the change (for example labeling change, updates of regulatory files, need to notify regulatory bodies, etc.).
- Verification activities to ensure compliance with product specifications and/ or unforseen adverse effect.
- Design Validation activities and human factor testing.
- The need to conduct the change as a design and development project.
- Installation process and procedures.
- Resources required for the implementation of the change such as time, cost, human resources.

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- Cost of the final product.
- Other as per the nature and the scope of the change.

5.4.2.10. Activities related to the implementation of the change:

- Software life cycle activities.
- Design Reviews.
- Update of documents such as work instructions, product specifications, etc.
- Supplier approval.
- Training of PeriGen personnel.
- User training.
- Service, installation and Marketing activities.
- Update of Labeling.
- Regulatory submissions.
- Field actions (update product version for current prodcut customers), Changes/ upgrades to products already placed in the market.
- Other as per the nature of the change.

5.4.2.11. Input shall be obtained from company personnel as well as other applicable sources (subcontractors, consultants, etc.) as required to determine the applicability and consequences of the proposed change.

5.4.2.12. The Management Representative is responsible to identify if the change is a significant change, and requires notification of regulatory bodies or a need to implement the Recall Procedures (**RA-WI-005**, *FDA Recall and Reporting Requirements*, **RA-WI-006**, *Canadian Recall and Reporting Requirements*) or any related field action.

5.4.2.13. The Product Owner or someone nominated by him and Management Representative shall record the required approval activities/ deliverables in the **FQMS-7.3.7-003**, *Engineering Change Order (ECO) Form*, including responsible person.

5.4.2.14. The SVP R&D shall review the results of the evaluation activities and decide if to accept, alter or reject the change according to the inputs provided by the reviewers. The decision shall be recorded and authorized in the **FQMS-7.3.7-003**, *ECO Form*.

5.4.2.15. The QA Manager will assign a number to each **FQMS-7.3.7-003**, *Engineering Change Order (ECO) Form*. The QA manager will record the

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ECO in the **FQMS-7.3.7-001,** *ECO Control Log*. The ECOs shall be numbered as follows:

XX-YYY: Where XX is the calendar year and YYY is sequential.

5.4.2.16. If the change is rejected, the reason for rejection is documented and the form is filed in the Design changes file.

5.4.2.17. If the change shall be handled as a design and development project the SVP R&D is responsible to nominate a Development Project Manager. The project will be handled in accordance to Design and Development Control procedures and **SW-WI-001,** *Software Development Life Cycle* procedure.

5.4.2.18. The change shall be implemented only after receiving all inputs from the assigned reviewers, and the approval of the **FQMS-7.3.7-003**, *Engineering Change Order (ECO) Form* by SVP R&D.

5.5. Reporting of Significant Changes

5.5.1. All design changes and all changes to the manufacturing process, critical subcontractors or the quality system shall be assessed in order to determine whether they need to be reported to Regulatory Authorities and/or to BSI, PeriGen's certification body (refer to Appendix A).

5.5.2. The results of this assessment shall be documented in the ECO form (**FQMS-7.3.7-003**, *Engineering Change Order (ECO) Form*) for product-related changes, Document Change Request (DCO) forms, or in regualtory folders, for QMS-related changes.

5.6. Change Implementation and Approval

5.6.1. All finalyzed and approved documentation will be submitted to the QA Manager who will monitor and control the process, gather all the documentation and update the **FQMS-7.3.7-003**, *ECO Form* accordingly. Whenever the Management Representative or QA Manager decides that some departmental involvement is not required, Management Representative will sign this function column with NA and sign with his initials.

5.6.2. The product can be released for marketing only after the succesful completion of all the activities/ deliverables required for the implementation of the change and aproval signature of the SVP R&D, Product Owner, Management Representative or QA Manager and if the change involves a software change by SPM (Software Project Manager).

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5.6.3. Notifications or submissions to pertinent regulatory authorities are performed according to the following procedures: **RA-WI-002,** *FDA Regulatory Requirements* and **RA-WI-003**, *Canadian Regulatory Requirements*.

5.6.4. All ECO forms (**FQMS-7.3.7-003**) shall be filled in the ECO file kept by the QA/RA department. A copy of (or reference to) an approved ECO shall be filed in the DHF.

5.6.5. The QA Manager is responsible to monitor the status of all open Engineering Change Orders in the company.

5.7. Change order distribution

5.7.1. The QA/ RA department is responsible to distribute the applicable documents among the applicable workgroups.

5.8. Field Change Order (FCO) process

Whenever an ECO calls for a Field Change Order, this will be identified in the **FQMS-7.3.7-004**, *Field Change Order (FCO) Form*. The FCO will then be prepared under the responsibility the Management Representative and forwarded to the Operations Department.

6. Responsibilities

6.1. The SVP R&D has the overall responsibility for implementing this procedure.

6.2. All personnel involved in the project are responsible, when needed, to initiate and process changes as described in this procedure.

6.3. Any employee who detects a need for change in a product, component, and/or process, following its release to market, is responsible to initiate a change request.

6.4. It is the responsibility of the Management Representative to ensure that the pertinent Regulatory Bodies and authorized agents including certification Body, FDA, Health Canada etc. are informed, when required, about changes in the company products and/or in related Quality Management System processes, as applicable.

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Appendix A – Significant Changes

According to ISO 13485 changes must be determined for their significance to function, performance, usability, safety and applicable regulatory requirements for the medical device and its intended use.

Changes that need to be reported to BSI:
BSI are involved in the process of certification of the Quality System in accordance with ISO 13485 and MDSAP, thus PeriGen has determined the changes that are required to be reported to BSI, including:

- Any plan of substantial changes to the quality system;
- Any plan of substantial changes to the product-range covered by the quality system;
- Any changes to the approved design that could affect the conditions prescribed for use of the device (i.e. the indications and/or contraindications determined by PeriGen to be appropriate to ensure safety and effectiveness for the intended use of the device).
- When considering whether or not a particular device change is 'substantial', the following should be taken into account:
  - Does the change affect the performance specification of the device?
  - Does the change affect the materials the device is composed of?
  - Does the change introduce new hazards which have not been previously addressed?
  - Does the change affect the risks associated with existing hazards?
  - Does the change alter the details on compliance with the essential requirements given in the Technical File?
  - Does the change alter the details on intended use given in the Technical File?
  - Does the change mean that the device will have different end users or be used in a different manner?
  - Does the change mean that the clinical data/ performance evaluation data for the original device is not sufficient to confirm conformity of the changed device with the required characteristics and performance?
  - Is the change a direct result of actions taken related to concerns arising from incidents/recalls/complaints?
- Changes to the quality system are considered 'substantial' if:
  - The change affects compliance of the devices covered by the quality system with the essential requirements or the approved type/design, or;
  - The change affects the compliance of the quality system with its own regulatory requirements.

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- When considering whether or not a particular quality system change is 'substantial', the following considerations should be taken into account:
  - Does the change affect PeriGen's ownership?
  - Does the change affect the location of PeriGen's activity?
  - Does the change affect the manufacturing technologies (processes and equipment)?
  - Does the change affect device conformity with the essential requirements or the approved type/design (e.g. change of supplier)?
  - Does the change affect the arrangements implemented to achieve continued compliance of the quality system with the relevant standards (e.g. design verification, design validation, organizational structure, processes interaction, quality control procedures, addition or deletion or move of a facility)?
  - Does the change affect the arrangements (e.g. verification, validation, organizational structure) for ensuring continued compliance with the requirements?
- A change to the product-range covered by the quality system is considered 'substantial' if it affects the categories of medical devices designed and/or manufactured through the quality system.
- Any change related to the terms of the contract between PeriGen and the certification body is a substantial change even if it is not directly related to conformity or performance of the medical device (e.g., change of the medical device identification, of business name, of the company's owner, etc.).

Changes to report to Health Canada & BSI:

Significant change means a change that could reasonably be expected to affect the safety or effectiveness of a medical device. It includes a change to any of the following:

- Substantial changes are made to a licensed device, for example changes made to the name of a licensed device, device manufacturer's name or address;
- the manufacturing process, facility or equipment;
- the manufacturing quality control procedures, including the methods, tests or procedures used to control the quality, purity and sterility of the device or of the materials used in its manufacture;
- the design of the device, including its performance characteristics, principles of operation and specifications of materials, energy source, software or accessories; and
- the intended use of the device, including any new or extended use, any addition or deletion of a contra-indication for the device, and any change to the period used to establish its expiry date.

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- When there is a new or modified ISO 13485 MDSAP certificate issued, the Management Representative is obligated to inform and submit the new or modified certified to Health Canada using the fax-back F202 form within 30 days of the certificate issue.

Changes to report to FDA:

- Changes in ownership, corporate or partnership structure, or location of the facility will be notified to FDA within 30 days of any change.
- Change in the US agent's name, address, or telephone number will be notified to FDA within 10 days of any change. Change in the Initial Importer's address, or telephone number will be notified to FDA at the time of any changes.
- For most types of devices, FDA requires the manufacturer to submit a premarket notification, or 510(k), submission at least 90 days before:
    - Introducing a new device for commercial distribution in the U.S.
    - Implementing any change or modification that could significantly affect the safety or effectiveness of an existing device.

  Considerations for when a change triggers a new submission shall be made according to **RA-WI-002** - *FDA Regulatory Requirements* procedure.

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